

Karman Power and Labour PyRoll Plugin

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The PyRoll plugin `pyroll-karman-power-and-labour` calculates the roll-force and roll-torque as well as the mean neutral plane location for a roll-pass. Since von-Karman originally developed the strip or slap theory for flat rolling, the grooved pass is not considered directly. The solution is therefore derived for an equivalent flat pass, which makes the plugin's accuracy heavily depended on the equivalent flat pass plugins.

The plugin provides four solvers, covering the full range from thick slabs to metal foils, and selects between them automatically for each roll pass:

- a rigid-roll solver based directly on von-Kármán's differential equation (section 1), valid whenever elastic effects are negligible;
- a rigid-plastic solver with a mixed Coulomb/sticking friction law (section 2), used for medium passes, that keeps von-Kármán's structure but replaces pure Coulomb friction with Bay and Wanheim's smoothed mixed law;
- an elastic-plastic layer model (section 3), used for thick passes, that adds elastic entry/exit zones, the same mixed friction law, through-thickness resolution into multiple layers, and full thermal coupling;
- a full elastic-contour foil-rolling model (section 4), used once elastic roll flattening becomes so severe that a circular roll-gap assumption is no longer valid.

section 5 explains the criteria used to select between the four, and section 7 lists the hooks the plugin defines. A fifth model, Orowan's own 1943 hot-rolling theory (section 6), is also implemented but deliberately kept outside this automatic selection - section 6 explains why and how it is used instead.

1 Rigid-roll model (von-Kármán)

The current model is based on the classic elementary theory of plasticity, which is a well known approach of modelling the stress distributions in flat rolling. The elementary

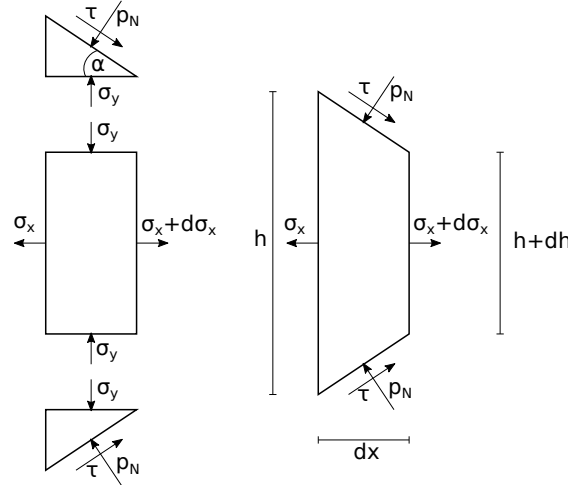


Figure 1: Stripe element of elementary theory of plasticity

theory of plasticity for flat rolling is also known as strip or slap theory, because the roll gap is divided in rolling direction in infinitesimal narrow strips, as shown in Figure 1, on which the force balance is built. In thickness direction the strip's height is equal the local height of the roll gap. The deformation of each strip is always plane and parallelepipedic. The strip's material properties are constant in thickness direction and may vary in rolling direction. Therefore, the vertical stress is constant in thickness direction and no shear deformation can be considered.

The roll surface is described as a circular arc, since elastic roll deformation is neglected in this model (see section 3 and section 4 for the elastic-plastic and elastic-contour models, which lift this restriction). So the roll gap height can be described by Equation 1. The rolls are here considered to be rigid. Also, the rolls are considered to be symmetric, meaning of equal radius R_W and rotational frequency n_W .

$$h(x) = s + 2 * \left(R_W - \sqrt{R_W^2 - x^2} \right) \quad (1)$$

The model needs additional approaches for the yield stress, and the friction between rolls and workpiece. The shear stress resulting from friction between the material and the roll surface is model by Coulomb's friction law. Since normally this model is used for cold rolling with lubricates, the friction coefficient μ has to be sufficiently high. The force balance at the stripe element leads to the well known Kármán differential equation [1, 2] as shown in Equation 4, which is an ordinary differential equation for the horizontal stress σ_x in x and can be solved easily by numerical methods. A suitable yield criterion gives the connection between σ_x and σ_y . In this approach the yield criterion according to von-Mises was chosen [3, 4].

$$0 = \sigma_x, \frac{dh}{dx} + h \frac{d\sigma_x}{dx} + 2\tau_R - 2p_N \tan(\alpha) \quad (2)$$

$$p_N = -\sigma_y - \tau_R \tan(\alpha) \quad (3)$$

$$\tau = \mu p_N \quad (4)$$

This rigid-roll solution never resolves elastic zones (the whole contact is assumed plastic from the first point of contact) and does not couple to temperature; it is implemented in `karman_solver.py` as `KarmanSolver`.

2 Mixed-friction model for medium passes

Pure Coulomb friction, integrated all the way across the contact as in Equation 4, demands ever-increasing shear traction toward the roll gap centre; once a pass is thick or high-friction enough, this exceeds what the material can actually transmit and produces an unrealistic, unbounded pressure spike. `karman_mixed_friction_solver.py` implements `KarmanMixedFrictionSolver`, which keeps von-Kármán's force balance exactly as in Equation 4, but caps the friction shear stress at the material's sticking (shear-yield) limit once Coulomb friction would exceed it, giving a friction hill that saturates ("sticking") rather than diverging, using the same smoothed Bay and Wanheim [5] mixed Coulomb/sticking law as the layer model (section 3, Equation 6), applied to the whole (single, homogeneous) cross-section rather than per layer.

Like `KarmanSolver`, this solver is rigid-plastic: it has no elastic entry/exit zones and no thermal coupling - those are a separate refinement, orthogonal to the friction-law refinement here, and are covered instead by the layer model once a pass is thick enough to need them (section 5).

This is *not* Orowan's own 1943 slab theory [6], despite addressing the same underlying observation (that pure Coulomb friction becomes unrealistic once passes are thick or high-friction enough). Orowan's actual model replaces the straight strip-element boundaries of Figure 1 with circular arcs, centred on the roll-gap symmetry line and meeting the roll surfaces at right angles, and corrects the horizontal force balance with an inhomogeneity function $\omega(a, \alpha)$ - an integral over the arc's polar angle - that captures genuinely non-uniform through-thickness deformation directly, rather than refining the friction law on top of the same homogeneous-strip geometry as section 1. Orowan himself noted the general theory was too complex for practical calculation at the time; later simplifications (e.g. Sims' application of it to hot rolling with sticking friction throughout) are reported to give unrealistic results (forming efficiencies above 100%) under some conditions. section 6 implements Orowan's own inhomogeneity-function theory (its sticking-friction branch) as a separate, explicitly-selected model, rather than as a further refinement dispatched automatically alongside the models above.

3 Elastic-plastic layer model for thick passes

For thick passes, the models above miss two further effects that become significant as the relative pass thickness grows: an elastic entry/compression zone and elastic recovery zone at exit (rather than the whole contact being plastic immediately), and, once the strip is thick enough relative to the contact length, meaningfully different flow stress between the surface (which alone feels roll-contact friction) and the core through the pass. `layer_solver.py` implements `LayerRollingSolver`, a layer-count-parametrized model following Weiner [7] (based on his *RollingHomogeneousElasticPlastic* and *RollingLayerModelElasticPlastic* implementations), which addresses both:

- the strip is discretized into N through-thickness layers (used here for $N > 1$, the "thick slab" regime; $N = 1$ collapses to a single homogeneous slab but is not used by this plugin's dispatch, since `KarmanMixedFrictionSolver` covers that case without elastic zones);
- each layer starts elastic and yields once its own local Mises criterion is met, giving genuine elastic entry and exit zones per layer instead of a single sharp elastic/plastic split;
- friction at the roll interface, and between adjacent layers, follows the same mixed Coulomb/sticking law as section 2 after Bay and Wanheim [5] (Equation 6), transitioning smoothly from Coulomb sliding to a sticking-limited shear stress as normal pressure grows;
- deformation heat, friction heat, conduction between neighbouring layers (and to the rolls), and convective transport are solved simultaneously with the stress equations, so each layer's temperature - and hence its flow stress - evolves individually through the pass.

$$p_{N,\text{Coulomb}} = \frac{-\sigma_y}{1 + \mu \tan(\alpha) \text{sign}}, \quad p_{N,\text{crit}} = \frac{m k_f}{\mu \sqrt{3}} \quad (5)$$

$$\tau = \left[\mu p_{N,\text{Coulomb}} (1 - \Phi) + \frac{m k_f}{\sqrt{3}} \Phi \right] \text{sign}, \quad \Phi = \frac{1}{\pi} \arctan \left(\frac{p_{N,\text{Coulomb}} - p_{N,\text{crit}}}{c p_{N,\text{crit}}} \right) + \frac{1}{2} \quad (6)$$

where m is the sticking friction coefficient (`friction_stiction_coefficient`), `sign` the (regularized) direction of relative sliding, and c a smoothing parameter (`pressure_smoothness`) controlling how sharply the transition Φ switches from Coulomb ($\Phi = 0$) to sticking ($\Phi = 1$) around the critical pressure $p_{N,\text{crit}}$.

Roll flattening is deliberately out of scope here, as it is for `KarmanSolver` and `FoilRollingSolver`'s own base radius: all three solvers simply read `Roll.working_radius` once rather than deriving a flattened radius themselves. A separate plugin providing a Hitchcock [8] (or other) flattened radius via that hook is therefore picked up transparently by every solver

in this plugin, with no special-casing needed; the classical linear Hitchcock relation implemented in `condition.py` (`hitchcock_radius_ratio`) exists solely to *classify* a pass for dispatch (section 5), and is never fed back into a calculation.

Because each pass only sees the incoming profile’s own scalar strain and temperature (`pyroll-core` has no native concept of a through-thickness state carried from one pass to the next), all layers start uniform; the elastic entry zone therefore has no mechanism to break symmetry between layers (matching the reference implementation, which defines but never uses a redundant-shear correction after Ekelund [9]). Layers still diverge from the first plastification onward, since only the outermost layers feel roll-contact friction directly while inner layers feel the weaker inner-friction law, and the resulting temperature and flow-stress differences compound through the rest of the pass.

4 Full elastic-contour foil-rolling model

Once elastic roll flattening becomes severe enough, even a scalar (Hitchcock) roll-flattening correction to `Roll.working_radius` is no longer accurate, because the flattened roll-gap shape stops being well approximated by a single effective radius. `foil_solver.py` implements `FoilRollingSolver`, following Mauk and Overhagen [10], which instead computes the true, generally non-circular, deformed roll-gap shape directly: the normal-pressure distribution along the contact is superposed with elastic half-space (Boussinesq/Johnson [11]) influence coefficients to obtain the roll surface’s own elastic deflection, iterated to convergence with the strip’s own elastic-plastic stress solution (elastic entry, plastic slip, an optional sticking zone, plastic slip, elastic exit).

5 Selecting between the four models

`roll_pass.py` wires the four solvers together behind a single hook, `karman_solution`, so that `roll_force`, `Roll.roll_torque`, `Roll.neutral_point` and the profile velocity hooks work unchanged regardless of which solver actually ran:

1. if the roll or profile elastic properties (`elastic_modulus`, `poissons_ratio`) are not set, the pass falls back to the rigid-roll model of section 1 unconditionally, since none of the elastic effects (or the Hitchcock-ratio/ L_d - H_m checks below) can be evaluated;
2. otherwise, the Hitchcock flattened-to-nominal radius ratio r'/r is evaluated (Mauk and Overhagen [10] give $r'/r \geq 2$ as the point where a circular roll-gap assumption is no longer trustworthy); if exceeded, the foil-rolling model of section 4 is used;
3. otherwise, the classical contact-length-to-mean-thickness ratio L_d/H_m decides between the layer model’s thick-slab configuration (section 3) and the mixed-friction model (section 2) for medium passes.

All three limits are themselves hooks (Table 3) and can be overridden per pass, including forcing a particular model unconditionally. `KarmanMixedFrictionSolver` does not

itself need the elastic properties or thermal data the other refinements do, but is kept behind the same elastic-properties gate as step 1 for consistency, rather than being dispatched independently of it.

Table 1 summarizes all five models by their underlying stress-strain treatment, friction law, and the thickness regime each is intended for - the four dispatched automatically by the criteria above, plus `OrowanSolver` (section 6), which is not.

6 Orowan's (1943) hot-rolling model

Every model in section 1–section 4 keeps von-Kármán's original strip geometry (Figure 1): straight, vertical strip boundaries, a homogeneous stress state through the strip's thickness at each x , and (in section 3) at best a coarse, discretized approximation of through-thickness variation via a handful of layers. Orowan's 1943 theory [6] - as reconstructed here from Overhagen's account of it [12, sec. 4.4.1.2], the original 1943 paper not being available for direct comparison - instead changes the strip element's own geometry: circular arcs, centred on the roll-gap symmetry line and meeting the roll surfaces at right angles, replace the straight vertical boundaries, with a polar coordinate system on each arc carrying an inner tangential stress $\sigma_{t,i}$, radial stress $\sigma_{r,i}$ and shear stress τ_i . This directly represents the non-uniform, redundant through-thickness deformation Orowan observed rolling plasticine specimens, rather than approximating it through a handful of discrete layers as section 3 does.

`orowan_solver.py` implements `OrowanSolver`: the horizontal force balance is von-Kármán's own equation, unchanged from Equation 4's first line, but the relation between σ_x and σ_y that closes it is no longer the plain Mises offset - it is Orowan's own closure, correcting for the inhomogeneous deformation the circular-arc geometry captures:

$$\sigma_x = \sigma_y + \frac{2k_f}{\sqrt{3}} \left[\omega_O(\alpha) \mp \frac{1}{2} \left(\frac{1}{\alpha} - \frac{1}{\tan \alpha} \right) \right] \quad (7)$$

$$\omega_O(\alpha) = \frac{1}{\sin \alpha} \int_0^\alpha \sqrt{1 - \left(\frac{\theta}{\alpha} \right)^2} \cos \theta \, d\theta = \frac{\pi J_1(\alpha)}{2 \sin \alpha} \quad (8)$$

where θ is the polar angle within an arc and J_1 the order-1 Bessel function of the first kind. The "-" sign in Equation 7 applies on the entry (Vorlauf) side, "+" on the exit (Nachlauf) side, mirroring the sign convention already used for τ_R in Equation 4. The inhomogeneity function ω_O ranges from 1 (frictionless limit) down to $\pi/4 \approx 0.785$ as $\alpha \rightarrow 0$ (full sticking) and depends only weakly on α otherwise [12, Abb. 4.4/4] - it is what makes Equation 7 a genuine correction to, rather than a mere re-derivation of, the plain Mises relation $\sigma_x = \sigma_y + 2k_f/\sqrt{3}$ every other model in this plugin uses. Orowan's general theory pairs Equation 7 with a sliding-friction counterpart (his Gl. 4.4/5) and a friction-ratio parameter $a_O = 2\mu\sigma_N/k_f \in [0, 1]$ blending between the two; `OrowanSolver`'s default, `friction_model="sticking"`, implements only the sticking-friction branch above ($a_O \equiv 1$) applied throughout the contact, since sticking friction (from scale, oxide layers, and high temperature) is the realistic assumption across the whole arc in hot rolling - the

Table 1: Overview of stress-strain and friction models, with recommended thickness regime.

Model	Stress-strain model	Friction model	Recommended regime
Karman-Solver	rigid-plastic, homogeneous cross-section	pure Coulomb	fallback only - used whenever <code>elastic_modulus/poissons_ratio</code> are not set, at any thickness; superseded by one of the other four the moment elastic data is available
Karman-Mixed-Friction-Solver	rigid-plastic, homogeneous cross-section	Bay/Wanheim mixed Coulomb/-sticking	medium gauge: $r'/r < 2$ (default) and $L_d/H_m \geq 1$ (default) - the plugin's default once elastic data is present and neither condition below triggers
Layer-Rolling-Solver	elastic-plastic, through-thickness layers, full thermal coupling	Bay/Wanheim mixed (roll contact & inter-layer)	thick slabs: $L_d/H_m < 1$ (default), i.e. contact length small relative to mean thickness, and $r'/r < 2$
Foil-Rolling-Solver	elastic-plastic, homogeneous cross-section, elastic (non-circular) roll-gap contour	Coulomb with optional sticking zone	foil rolling: $r'/r \geq 2$ (default). Governed by r'/r , not gauge alone - force and roll stiffness matter too, so no fixed absolute thickness applies (our own 0.5 mm-to-0.2 mm steel-foil case needed a small roll radius and tuned friction/tension to actually reach $r'/r \geq 2$; industrial foils are typically well under 0.1 mm)
Orowan-Solver	elastic-plastic, single homogeneous cross-section, circular-arc inhomogeneity correction	sticking (default) or Bay/Wanheim mixed	not gated by L_d/H_m or r'/r - always selected explicitly (section 7). A hot-rolling alternative to <code>KarmanMixedFrictionSolver/LayerRollingSolver</code> , typically overlapping the medium-to-thick gauge range; default <code>friction_model="sticking"</code> for scale/oxide-dominated contacts, <code>"mixed"</code> otherwise

regime this model targets. Elastic entry and exit zones bound the sticking-plastic zone on both sides, using the same Hooke's-law construction as section 3's $N = 1$ case, so that (unlike Orowan's own rigid-plastic 1943 paper) the whole contact is not assumed plastic from first contact.

Reconstructing Equation 7–Equation 8 from [12] required three specific interpretive choices, each checked as far as the available material allows but not verifiable against Orowan's original 1943 paper directly - documented in full in `orowan_solver.py`'s module docstring, and summarized here:

- Overhagen's own Gl. 4.4/7 prints ω_O 's integration limit as a_O rather than α ; Equation 8 integrates to α instead, since only that choice reproduces the two values his own Abb. 4.4/4 plots ($\omega_O \rightarrow 1$ as $a_O \rightarrow 0$; $\omega_O \rightarrow \pi/4$ as $a_O \rightarrow 1, \alpha \rightarrow 0$ - both checked against Equation 8's closed form in `tests/test_orowan_solve.py`), and the a_O -limit version is not even well-defined in general (its square root goes negative once $a_O/\alpha > 1$, the common case at realistic rolling angles and high a_O);
- k_f in Equation 7 is $2\times$ the uniaxial flow stress `flow_stress_function` returns elsewhere in this plugin (Overhagen's own source equations use a bare $k_f/\sqrt{3}$ offset, consistent with an older, Tresca-flavoured plane-strain convention for k_f predating this plugin's Mises one) - chosen specifically so Equation 7 reduces to the ordinary Mises relation in the frictionless limit, keeping the elastic-zone yield-onset criterion (still the plain Mises condition) and the plastic-zone closure continuous with each other;
- the entry/exit sign of the $(\frac{1}{\alpha} - \frac{1}{\tan \alpha})$ term is a best-effort assignment: Overhagen's source prints it as " $\mp$ " without enough surrounding context in the excerpt available to fix which sign belongs to which side independently.

None of these affect ω_O 's qualitative role or Equation 7's overall structure; they are exactly the kind of detail worth re-checking against Orowan's 1943 paper directly if it becomes available.

Passing `friction_model="mixed"` replaces the sticking-only branch above with Bay and Wanheim's mixed Coulomb/sticking law [5] (Equation 6, the same law section 2 and section 3 use) - for passes where friction is not high enough for sticking to hold across the whole contact, rather than Orowan's own a_O -dependent sliding branch (his Gl. 4.4/5). Rather than rederiving Gl. 4.4/5, this keeps Equation 7's structure unchanged and generalizes only ω_O : the sticking value $\omega_O(\alpha)$ of Equation 8 is linearly blended toward the frictionless limit 1 using Φ , the same smooth Coulomb-to-sticking transition variable Equation 6 already computes for the friction law itself,

$$\omega_{O,\text{eff}}(\alpha, \Phi) = (1 - \Phi) \cdot 1 + \Phi \cdot \omega_O(\alpha) \quad (9)$$

which is a deliberate, pragmatic hybrid rather than a literal implementation of Gl. 4.4/5 - justified by ω_O 's own weak dependence on its arguments (noted above), so linearly interpolating between its two known closed-form endpoints, weighted by the regime indicator already driving the friction law, avoids needing the general, quadrature-only $\omega_O(\alpha, a_O)$

Table 2: Orowan’s model compared to the plugin’s other solvers.

Model	Strip geometry	Friction law	Elastic zones	Thermal
KarmanSolver	straight, homogeneous	pure Coulomb	none	none
KarmanMixed-FrictionSolver	straight, homogeneous	Bay/Wanheim mixed	none	none
LayerRolling-Solver	straight, N discrete layers	Bay/Wanheim mixed	entry & exit, per layer	full
FoilRollingSolver	straight, homogeneous, elastic roll contour	Coulomb + optional sticking	entry & exit	none
OrowanSolver	circular arc, continuous inhomogeneity	sticking (defect) or Bay/Wanheim mixed	entry & exit	none

for $a_O \notin \{0,1\}$. Since Φ itself depends on σ_y - the quantity Equation 7 solves for - **OrowanSolver** iterates Equation 7 and Equation 9 to a fixed point (converging in a handful of iterations, given how weakly ω_O varies). Unlike the sticking-only default, whose elastic zones carry no friction at all (Bay and Wanheim’s law is well-behaved as pressure $\rightarrow 0$, unlike assuming sticking friction where contact pressure is still essentially zero - see `orowan_solver.py`’s module docstring), `friction_model="mixed"` applies the same mixed friction law in the elastic entry/exit zones too, exactly as section 3 already does.

OrowanSolver is not wired into the `karman_solution` dispatch hook of section 5: it must be instantiated directly, `OrowanSolver(roll_pass=roll_pass)`, for a pass whose profile has already been solved. This is a deliberate scope decision, not an oversight - the model is a genuinely different, orthogonal refinement (curved-strip inhomogeneity) from the axis section 5 already selects along (elastic flattening, then contact-length-to-thickness), it applies only to a single homogeneous cross-section (no layer-count parametrization, unlike section 3), and it has no thermal coupling (flow stress is evaluated at the incoming profile’s own scalar temperature throughout, like section 1 and section 2). Table 2 summarizes how it relates to the other models in this plugin.

7 Usage instructions

The plugin can be loaded under the name `pyroll_karman_power_and_labour`.

An implementation of the `roll_force` hook on `RollPass` is provided. Furthermore, an implementation of the `roll_torque` and `neutral_point` hooks on `RollPass.Roll` is provided.

Table 3: Hooks specified by this plugin.

Hook name	Meaning
<code>coulomb_friction_coefficient</code>	Coulomb's friction coefficient μ
<code>back_tension</code>	Back tension of the roll pass $\sigma_{x,0}$
<code>front_tension</code>	Front tension of the roll pass $\sigma_{x,1}$
<code>karman_solution</code>	Solver object (of one of the four models) with solution values as attributes
<code>foil_rolling_condition</code>	Whether the foil-rolling model (section 4) is used
<code>foil_rolling_hitchcock_limit</code>	Hitchcock ratio r'/r above which the foil model is used (default 2.0)
<code>thick_slab_condition</code>	Whether the multi-layer, thick-slab configuration (section 3) is used instead of <code>KarmanMixedFrictionSolver</code> (section 2)
<code>layer_model_ld_hm_limit</code>	L_d/H_m ratio below which the multi-layer configuration is used (default 1.0)
<code>layer_model_layer_count</code>	Number of through-thickness layers on the thick-slab path (default 5)
<code>friction_stiction_coefficient</code>	Sticking friction coefficient m feeding the mixed friction law (Equation 6) of <code>KarmanMixedFrictionSolver</code> and <code>LayerRollingSolver</code> (default 0.8)
<code>roll_heat_transfer_coefficient</code>	Heat transfer coefficient at the roll/material interface (default 6000 W/(m ² K))

Additionally, hooks on `RollPass` are defined, which are used in the calculation, as listed in Table 3. The hooks `back_tension`, `front_tension` and `coulomb_friction_coefficient` have to be set and adjusted individually; the foil and layer models additionally need `elastic_modulus` and `poissons_ratio` set on both `RollPass.Roll` and `RollPass.InProfile` to be selected at all (see section 5), and the layer model's thermal coupling needs `specific_heat_capacity`, `thermal_conductivity` and `density` set on both.

`OrowanSolver` (section 6) is not reachable through `karman_solution`, since it is not part of the automatic dispatch of section 5; call it directly once the pass has been solved, e.g. `OrowanSolver(roll_pass=roll_pass)` after `PassSequence([roll_pass]).solve(in_profile)`, then read `roll_force_per_unit_width`, `roll_torque_per_unit_width`, `neutral_plane_position` and `solution` off the returned instance directly, the same attributes the other four solvers expose.

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